pass filter and sample the data at 64 kHz. The resulting digital data contains the desired voice band between 100 and 3000 hertz, but also has an unusable band between 3 kHz and 32 kHz. Second, remove these unusable frequencies in *software*, by using a *digital* low-pass filter at 3 kHz. Third, resample the digital signal from 64 kHz to 8 kHz by simply discarding every seven out of eight samples, a procedure called **decimation.** The resulting digital data is equivalent to that produced by aggressive analog filtering and direct 8 kHz sampling.

Multirate techniques can also be used in the output portion of our example system. The 8 kHz data is pulled from memory and converted to a 64 kHz sampling rate, a procedure called **interpolation**. This involves placing seven samples, with a value of zero, between each of the samples obtained from memory. The resulting signal is a digital *impulse train*, containing the desired voice band between 100 and 3000 hertz, plus spectral duplications between 3 kHz and 32 kHz. Refer back to Figs. 3-6 a&b to understand why this it true. Everything above 3 kHz is then removed with a *digital* low-pass filter. After conversion to an analog signal through a DAC, a simple RC network is all that is required to produce the final voice signal.

Multirate data conversion is valuable for two reasons: (1) it replaces analog components with software, a clear economic advantage in mass-produced products, and (2) it can achieve higher levels of performance in critical applications. For example, compact disc audio systems use techniques of this type to achieve the best possible sound quality. This increased performance is a result of replacing analog components (1% precision), with digital algorithms (0.0001% precision from round-off error). As discussed in upcoming chapters, digital filters outperform analog filters by *hundreds of times* in key areas.

Single Bit Data Conversion

A popular technique in telecommunications and high fidelity music reproduction is **single bit ADC and DAC**. These are multirate techniques where a higher sampling rate is traded for a lower number of bits. In the extreme, only a single bit is needed for each sample. While there are many different circuit configurations, most are based on the use of **delta modulation**. Three example circuits will be presented to give you a flavor of the field. All of these circuits are implemented in IC's, so don't worry where all of the individual transistors and op amps should go. No one is going to ask you to build one of these circuits from basic components.

Figure 3-16 shows the block diagram of a typical delta modulator. The analog input is a voice signal with an amplitude of a few volts, while the output signal is a stream of digital ones and zeros. A comparator decides which has the greater voltage, the incoming analog signal, or the voltage stored on the capacitor. This decision, in the form of a digital one or zero, is applied to the input of the latch. At each clock pulse, typically at a few hundred kilohertz, the latch transfers whatever digital state appears on its